Analysis of PDEs

Shing Tak Lam

October 5, 2023

1 Basics

Let $\mathcal{U} \subseteq \mathbb{R}^n$ be an open set. A *PDE of order k* is an expression of the following form

$$F(x, u, Du, \dots, D^k u) = 0 \tag{1}$$

where $u : \mathcal{U} \to \mathbb{R}$ is the unknown, $F : \mathcal{U} \times \mathbb{R} \times \mathbb{R}^n \times \cdots \times \mathbb{R}^{n^k}$ is a general function. We say that u is a *classical solution of* eq. (1) if u satisfies eq. (1) in $\mathcal{U}$.

We may also consider the case where $u(x) \in \mathbb{R}^p$ and $F \in \mathbb{R}^q$, and we call this a *system of PDEs*.

1.1 Examples of PDEs

Example 1.1 (ODE system)

One example of a model used in mathematical biology is the ODE system

$$\begin{aligned} \frac{\mathrm{d}u}{\mathrm{d}t} &= f(u, v) \\ \frac{\mathrm{d}v}{\mathrm{d}t} &= g(u, v) \end{aligned}$$

Example 1.2 (Laplace's equation)

$$\Delta u = \sum_{i=1}^{n} \partial_i^2 u = 0$$

The Laplacian is an *averaging* operator.

Example 1.3 (Heat equation)

$$u_t = D\Delta u$$

This is also called the diffusion equation, D is called the diffusion constant.

Example 1.4 (Navier-Stokes)

The Navier-Stokes equations in fluid dynamics is

$$\begin{aligned} u_t &= \nu \Delta u - u \cdot \nabla u - \nabla \rho + f \\ \operatorname{div}(u) &= 0 \end{aligned}$$

Example 1.5 (Transport equation)
The transport equation is

$$u_t + v u_x = 0$$

where v is a constant, corresponding to the velocity. A modification is the *advection-diffusion* equation,

$$u_t + v \cdot \nabla u = D\Delta u + f$$

Example 1.6 (Poisson equation)

$$\Delta u = f$$

Describes electric field due to some charge, or Newtonian gravity.

Example 1.7 (Wave equation)

$$\Box u = -u_t t + c^2 \Delta u = 0$$

This models sound waves, seismic waves, ...

Example 1.8 (KdV equation)
This equation admits *soliton* solution.

$$u_t + \partial_x^3 u - 6u\partial_x u = 0$$

Example 1.9 (Maxwell equations)

$$\begin{aligned} \operatorname{div}(E) &= \rho \\ \operatorname{div}(B) &= 0 \\ \partial_t E &= \nabla \times B = J \\ \partial_t B &= -\nabla \times F \end{aligned}$$

Example 1.10 (Einstein's equations)

$$\operatorname{Ric}(g) - \frac{1}{2} g R(g) = 0$$

1.2 Data and Well-Posedness

All of the examples from above need additional information to solve, which we call the *data*. For example, we might need $u|_{\partial \mathcal{U}}$ and so on. A guiding principle to this process is called *well-posedness (in the sense of Hadamard)*.

We say that a PDE problem (equation and the data) is well-posed if we have

1. A solution exists (in some function space).
2. Given some data, the solution should be unique (depends on the function space of choice).
3. The solution depends continuously on the data.

The aim is to find the largest space for which a solution exists, but small enough so that it is unique. (For example, strong cosmic censorship in GR?)

Notation 1.11 (Multi-index notation)**.** We will use multi-index notation,

$$\mathbb{N} = \{0, 1, \dots\}$$

$\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ is called a *multi-index*, and we define the *order* of α

$$|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$$

and the α-th derivative is

$$D^\alpha f(x) = \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n} f$$

If $x = (x_1, \dots, x_n)$, then $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$, and

$$\alpha! = \alpha_1! \cdots \alpha_n!$$

1.3 Classifying PDEs (of order k)

We say eq. (1) is *linear* if F is a linear function of x, u and its derivative. That is, we can write it as

$$\sum_{|\alpha| \le k} a_\alpha(x) D^\alpha u = f(x)$$

We say eq. (1) is *semilinear* if the highest order derivatives appear linearly with coefficients depending only on x. That is, we have

$$\sum_{|\alpha| = k} a_\alpha(x) D^\alpha u(x) + F(x, u, Du, \dots, D^{k-1}u) = 0$$

One example would be $\Delta u = u_x^2$. Finally, we say eq. (1) is *quasilinear* if the highest order derivatives appear linearly, but the coefficients depending on lower order derivatives. That is, we have

$$\sum_{|\alpha| = k} a_\alpha(x, u, \dots, D^{k-1}u) D^k u + F(x, u, \dots, D^{k-1}u) = 0$$

For example, we can have $uu_{xx} + u_{yy} - u_x^2 = 0$. Finally, we say eq. (1) is *fully non-linear* if it is non of the above.

2 Cauchy-Kovelevskaya theorem

2.1 ODE theory

Fix $\mathcal{U} \subseteq \mathbb{R}^n$ open, and suppose $f : \mathcal{U} \to \mathbb{R}^n$ is given. We would like to consider the ODE

$$\begin{cases} \dot{u}(t) = f(u(t)) \\ u(0) = u_0 \end{cases} \tag{2}$$

Theorem 2.1 (Picard-Lindelöf)**.** Suppose we have $r, K > 0$ such that $B_r(u_0) \subseteq U$ and

$$\|f(x) - f(y)\| \le K\|x - y\|$$

for all $x, y \in B_r(u_0)$. Then there exists $\varepsilon > 0$, depending on K, r and a unique C^1 solution $u : (-\varepsilon, \varepsilon) \to \mathcal{U}$ solving eq. (2).

3 Sobolev spaces

4 Elliptic PDEs

5 Hyperbolic PDEs